Design and Realization of an Affordable Continuous Sub-Cutaneous Insulin Infusion Pump for Precise Insulin Delivery

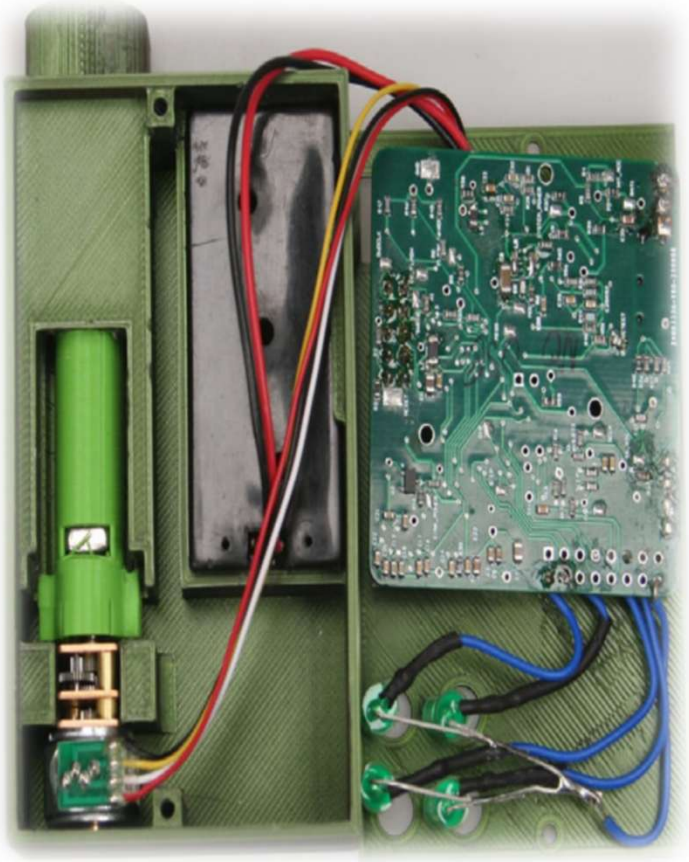
Team Members:

- Debjyoti Saha
- Eppa Manipreetham
- Harshaditya Punera

Project Mentor: Dr. Anirudh Nath



Introduction



**Matthew Payne-Design of an open source ultra low cost insulin pump*

The Clinical Challenge

Diabetes: A Global Crisis

- Type 1 Diabetes: Autoimmune destruction of pancreatic beta cells requiring lifelong insulin replacement
- Type 2 Diabetes: Progressive insulin resistance and declining beta-cell function
- 537 million adults worldwide affected, projected to reach 783 million by 2045

Risks of Poor Glycaemic Control

- **Hypoglycaemia:** Blood glucose <3.9 mmol/L causing confusion, seizures, unconsciousness
- **Hyperglycaemia:** Chronic elevation leading to retinopathy, nephropathy, neuropathy, cardiovascular disease
- **Precision imperative:** Tight glycaemic control (HbA1c <7%) reduces complications by 40-75%

Evolution of Insulin Delivery Systems

1920s-1980s

Syringes & Vials: Manual injection, poor precision, patient non-compliance, painful administration

1

1985-2000

Insulin Pens: Improved convenience and discrete dosing, but limited basal-bolus flexibility

2

2000-Present

Insulin Pumps: Continuous subcutaneous infusion, programmable basal rates, precise bolus delivery

3

Future

Closed-Loop Systems: Integration with continuous glucose monitoring for automated insulin delivery

4

Limitations of Manual Injection

Dawn Phenomenon: Early morning hyperglycaemia requires precise basal adjustment impossible with long-acting insulin

Compliance Burden: Multiple daily injections (4-6/day) reduce adherence, particularly in paediatric and geriatric populations

Dosing Inaccuracy: Pen increments of 0.5-1 IU insufficient for precise micro-dosing in children and insulin-sensitive patients



Insulin Dosing Principles

Insulin Types

- **Rapid-acting:** Onset 10-15 min, peak 1-2 hrs (lispro, aspart)
- **Short-acting:** Onset 30 min, peak 2-4 hrs (regular insulin)
- **Long-acting:** Duration 18-24+ hrs (glargine, degludec)

Concentration Standards *(IU=Insulin Units) (1 IU = 0.0347 mg)*

- **U-100:** 100 IU/mL (standard)
- **U-200:** 200 IU/mL (concentrated formulations)
- **Critical distinction:** Volume ≠ Units. Precision requires µL-level control

Basal Insulin Delivery

Purpose: Physiologic mimicry of pancreatic background insulin secretion (0.5-1.0 IU/hr typical range)

Requirements: Programmable hourly rates to address circadian variation. Precision: ±0.025 IU/hr minimum

Bolus Insulin Delivery

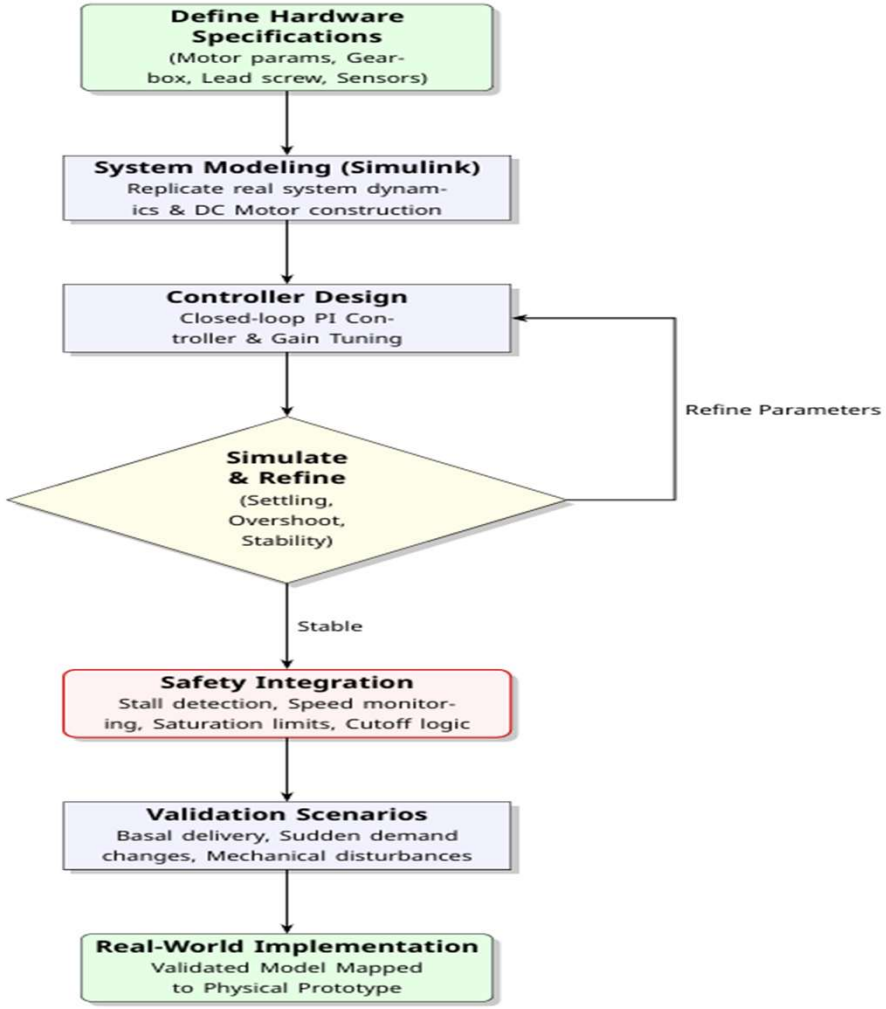
Purpose: Meal coverage and hyperglycaemia correction

Components: Carbohydrate ratio (1 IU per 10-15g CHO) plus correction factor (1 IU reduces glucose by 2-3 mmol/L)

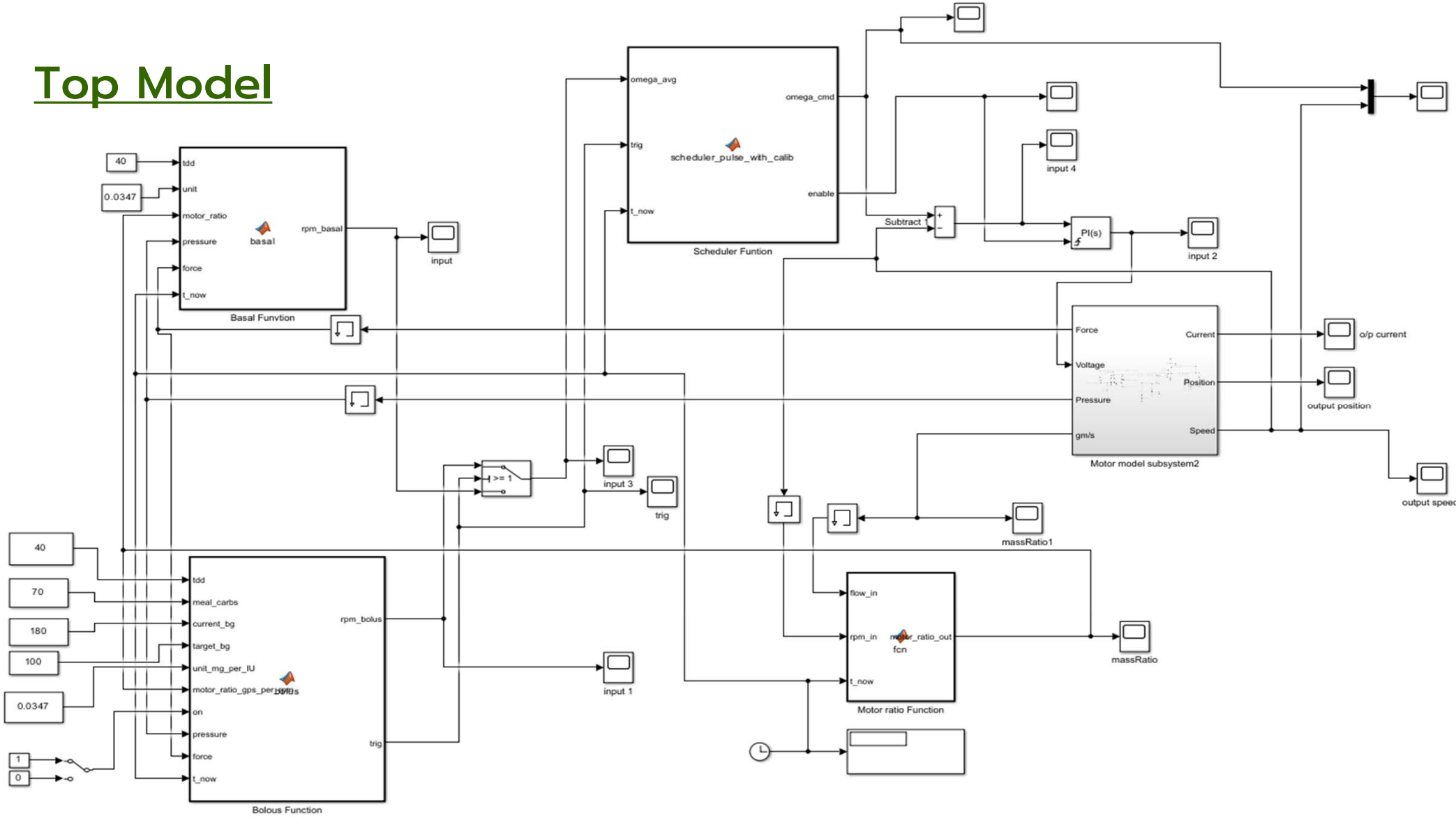
Requirements: Rapid delivery (0.05-15 IU), increment precision of 0.025 IU

Our Approach

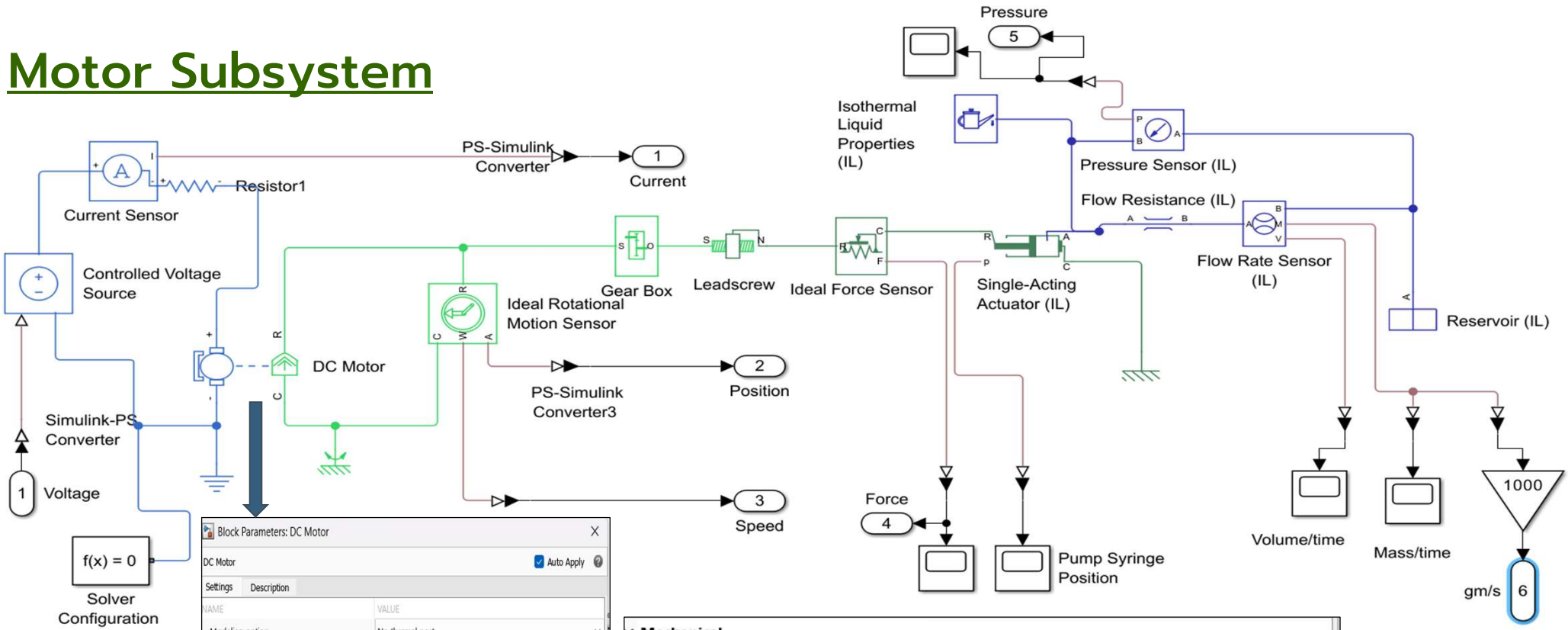
- Our approach is centered on developing a safe and high-precision insulin infusion pump using **MATLAB/Simulink**, with the simulation closely matching the actual hardware specifications that we will eventually use.
- We begin by modelling the complete system using the same hardware configuration that will be used in the final pump
- A model of the DC Motor is constructed, followed by the design of a closed-loop **PI controller** for precise speed regulation.
- To ensure patient safety, multiple protection layers are integrated into the model
- Finally, the integrated model is validated under various physiological and load scenarios such as basal delivery, sudden demand changes, and mechanical disturbances.

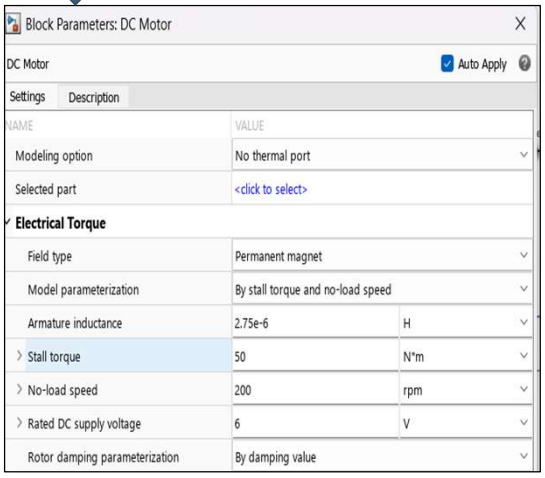


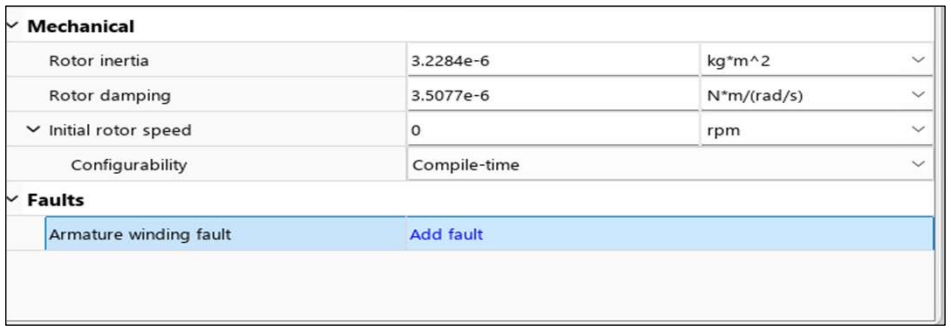
Top Model


40
0.0347
tdd
unit
motor_ratio
pressure
force
t_now
basal
rpm_basal
Basal Funvtion
input
omega_avg
trig
t_now
scheduler_pulse_with_calib
omega_cmd
enable
Scheduler Funtion
Subtract 1
PI(s)
input 2
input 4
Force
Voltage
Pressure
gm/s
Current
Position
Speed
Motor model subsystem2
o/p current
output position
output speed
>= 1
input 3
trig
massRatio1
40
70
180
100
0.0347
1
0
tdd
meal_carbs
current_bg
target_bg
unit_mg_per_IU
motor_ratio_gps_per_IU
on
pressure
force
t_now
bolus
rpm_bolus
trig
Bolous Function
input 1
flow_in
rpm_in
motor_ratio_out
fcn
t_now
Motor ratio Function
massRatio

Motor Subsystem







MATLAB Functions

1: Motor Ratio

Motor Ratio indicates how much fluid is delivered per unit motor speed:

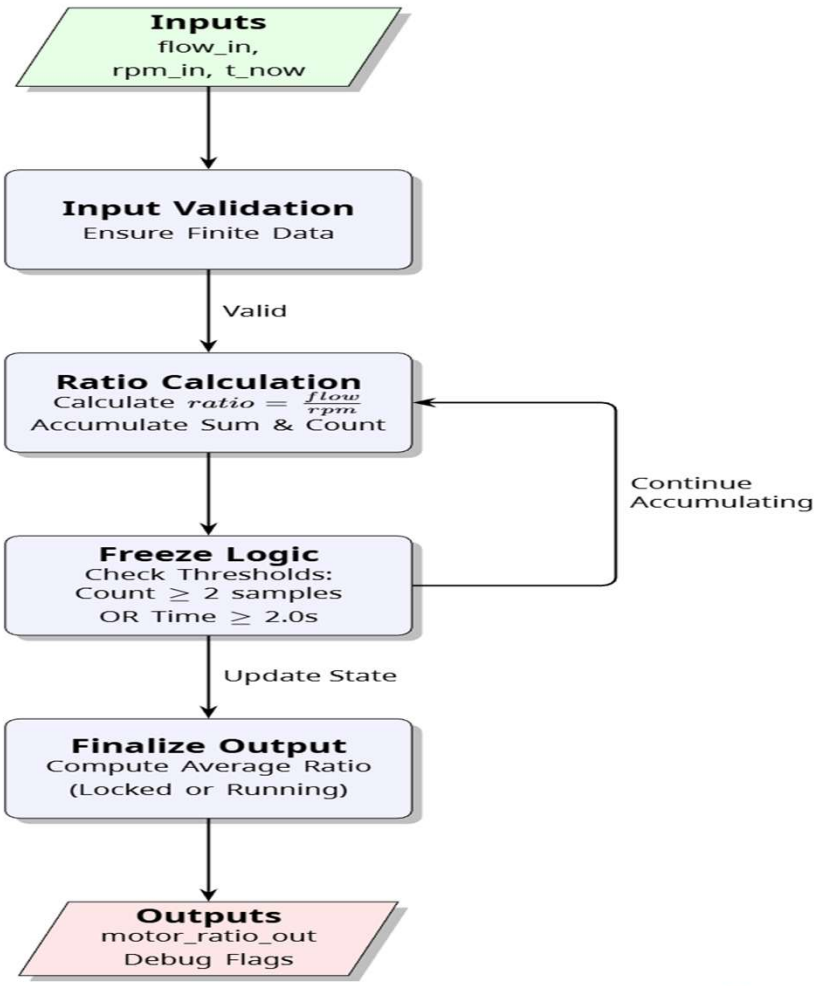
$$\text{Ratio} = \frac{\textit{Flow Rate}}{\textit{Motor } rad/s}$$

Working:

Input Validation:
Checks Flow, speed, and Time for valid values; if invalid → output 0.

Data Collection:
Once running, computes instantaneous ratio = Flow/rad/s and accumulates samples.

- **Fast Freeze (Sample-Based):**
  If enough valid samples (e.g., 2) are collected, it averages and freezes the value.
- **Time-Out Freeze:**
  If sample count is low but time limit (2 s) is reached, it averages collected data and freezes.



MATLAB Functions

2: Bolus Function

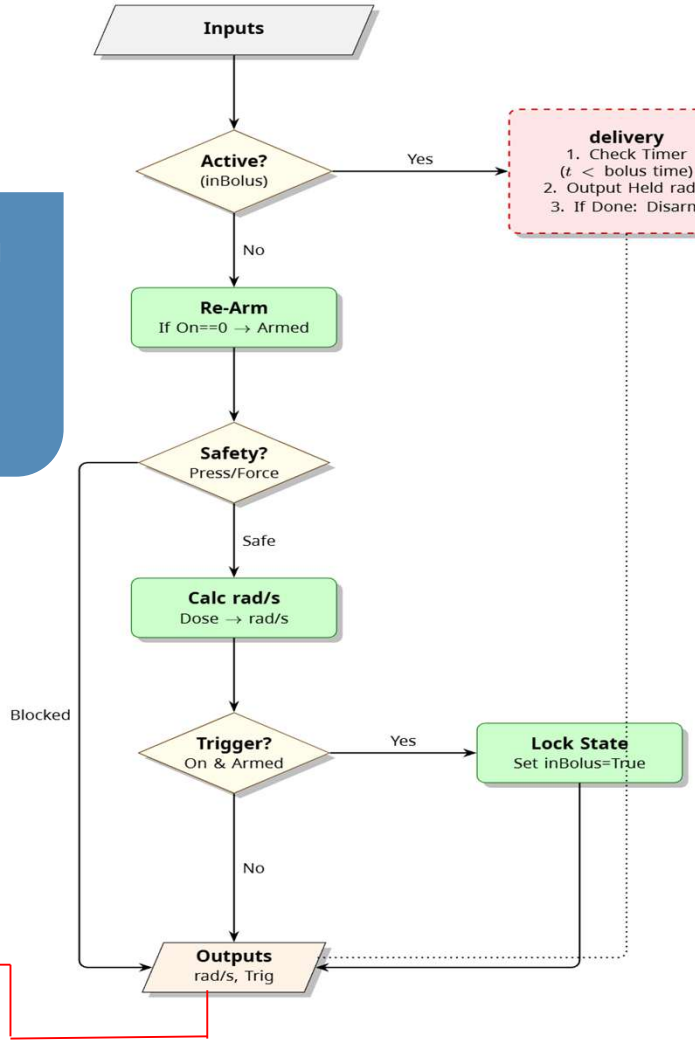
Inputs:

- Amount of carbohydrates that must be covered by the bolus.
- Patient's current and desired blood glucose level.
- Motor Ratio.
- Bolus On/Off.
- Feedback of Syringe Force and Pressure.

Compute Motor Speed (rad/s)

$$\omega = \left(\frac{Flow\ Rate}{Motor\ Ratio \times 1000}\right)$$

Required motor speed to deliver the bolus dose in given time. Sets **trig=1** indicating bolus delivery is ON.



Total Dose(Prescribed by Doctor)

$$\text{Total } Dose\ (IU) = \left(\frac{Carbs}{\frac{500}{Daily\ Dose}}\right) + \left(\frac{Current\ BG - Target\ BG}{\frac{1800}{Daily\ Dose}}\right)$$

Required Flow Rate (mg/s):
To deliver the Insulin in Bolus Time (here 5min.)

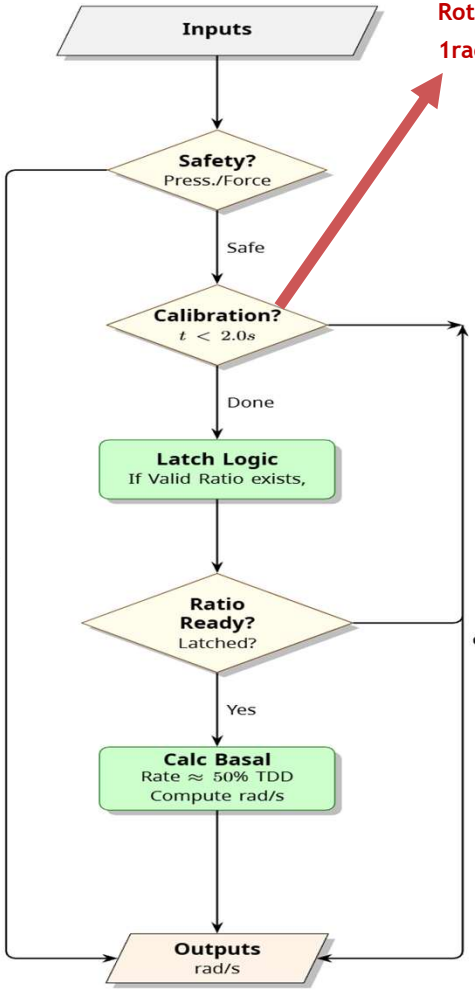
$$Flow\ Rate = \frac{Dose \times Insulin\ Mass\ per\ IU}{300}$$

MATLAB Functions

2: Basal Function

Inputs:

- Total daily insulin requirement.
- Insulin concentration factor
- Feedback of Syringe Force and Pressure.



One time calibration
Rotate motor for 2sec a
1rad/s to calc. Motor Ratio.

Basal Rate (Units/hr)

$$\text{Basal } Rate = \frac{0.5 \times Daily\ Dose}{24}$$

Required Mass Flow Rate (mg/s)

$$\text{Mass } Flow = \frac{Basal\ Rate \times Insulin\ Mass\ per\ IU}{3600}$$

Compute Motor Speed (rad/s)

$$\omega = \left(\frac{Flow\ Rate}{Motor\ Ratio \times 1000}\right)$$

Required motor speed to deliver the basal dose for 24 hours.

Problems:

1: The required motor speed is very low

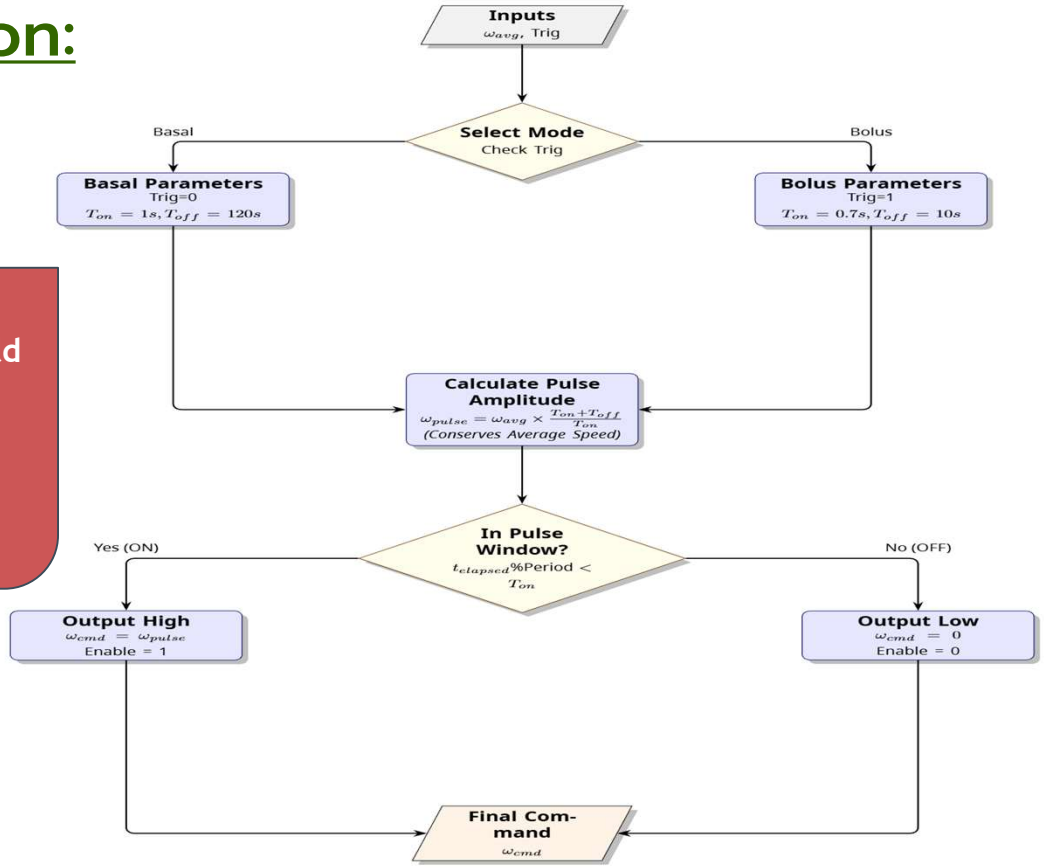
Because insulin doses are extremely small and the pump mechanism is highly geared, the motor only needs to rotate in order of about ***10^{-3} rad/s*** *to deliver the correct amount of insulin — ensuring ultra-precise, safe infusion.*

Solution:

Scheduling the doses

Volume delivered during the short ON-time = Volume that would have been delivered at the average speed continuously.

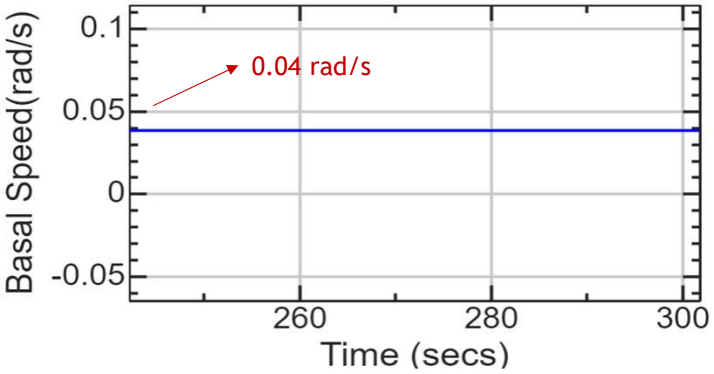
$$\text{Pulse } Speed = Average\ Speed \times \left(\frac{Total\ Cycle\ Time}{ON\ Time}\right)$$



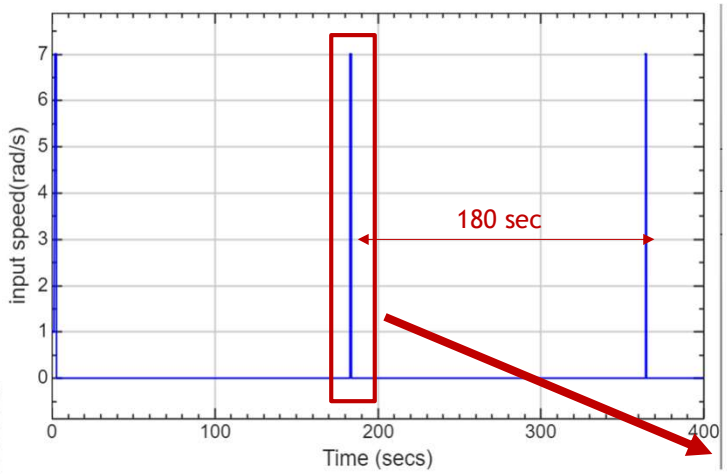
Enable will be used to control the working of PID controller.

Solution:

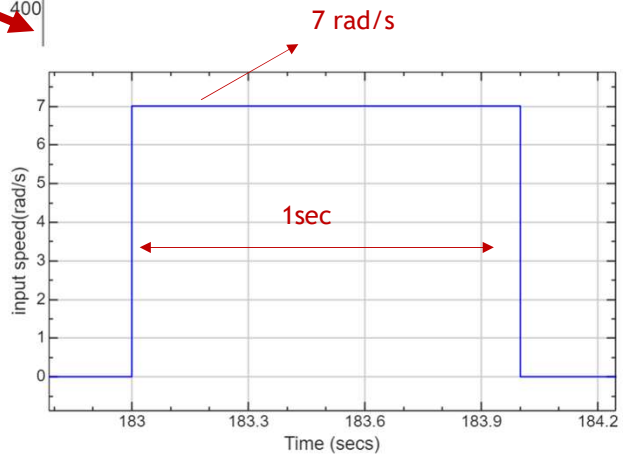
Scheduling the doses



Input to Schedular



Output of Schedular

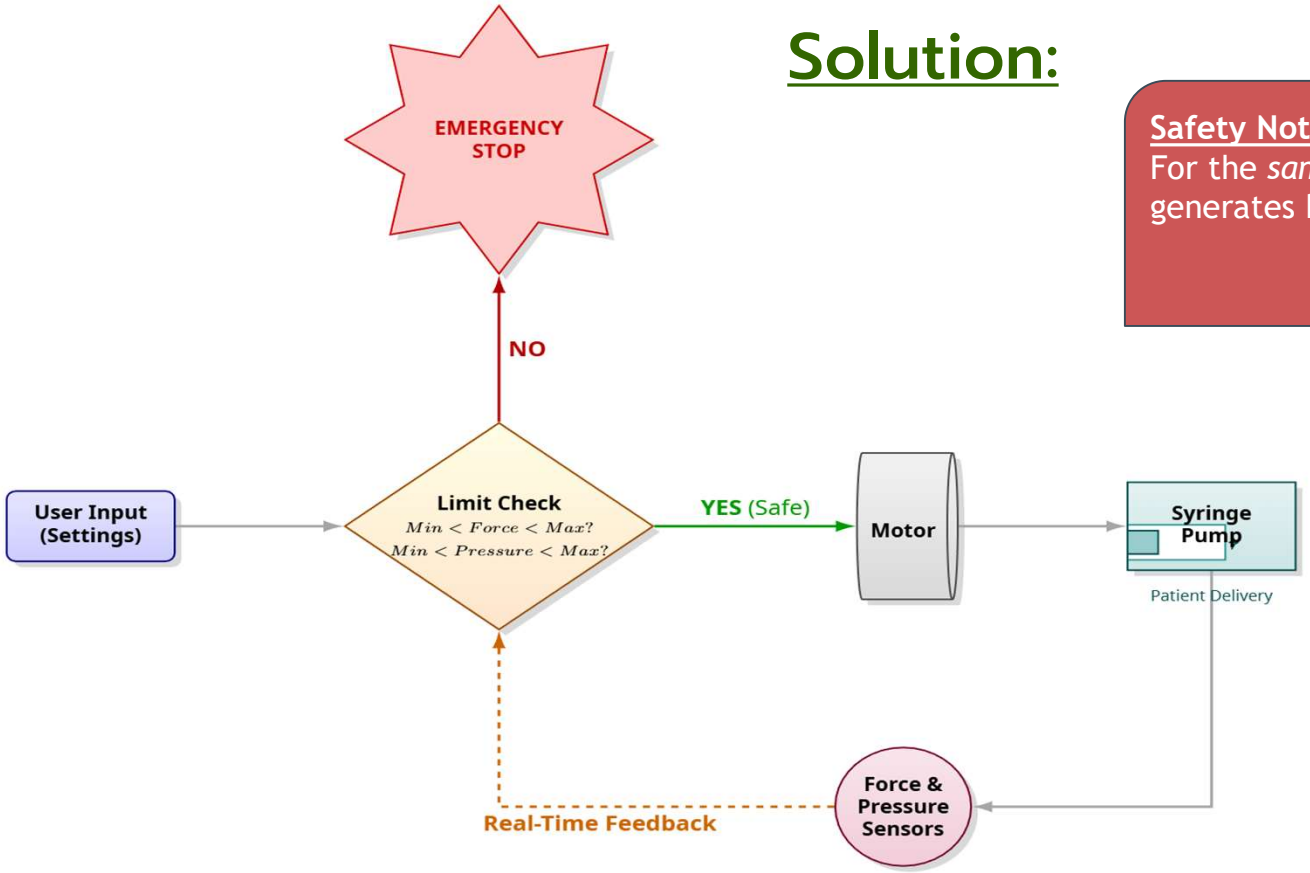


Problems:

2: Very high force/pressure exerted by piston/syringe

The motor may generate excessive force and pressure within the syringe due to computational errors or incorrect user inputs.

Solution:



Safety Note:
For the *same* motor force, a **smaller syringe** generates **higher pressure**.

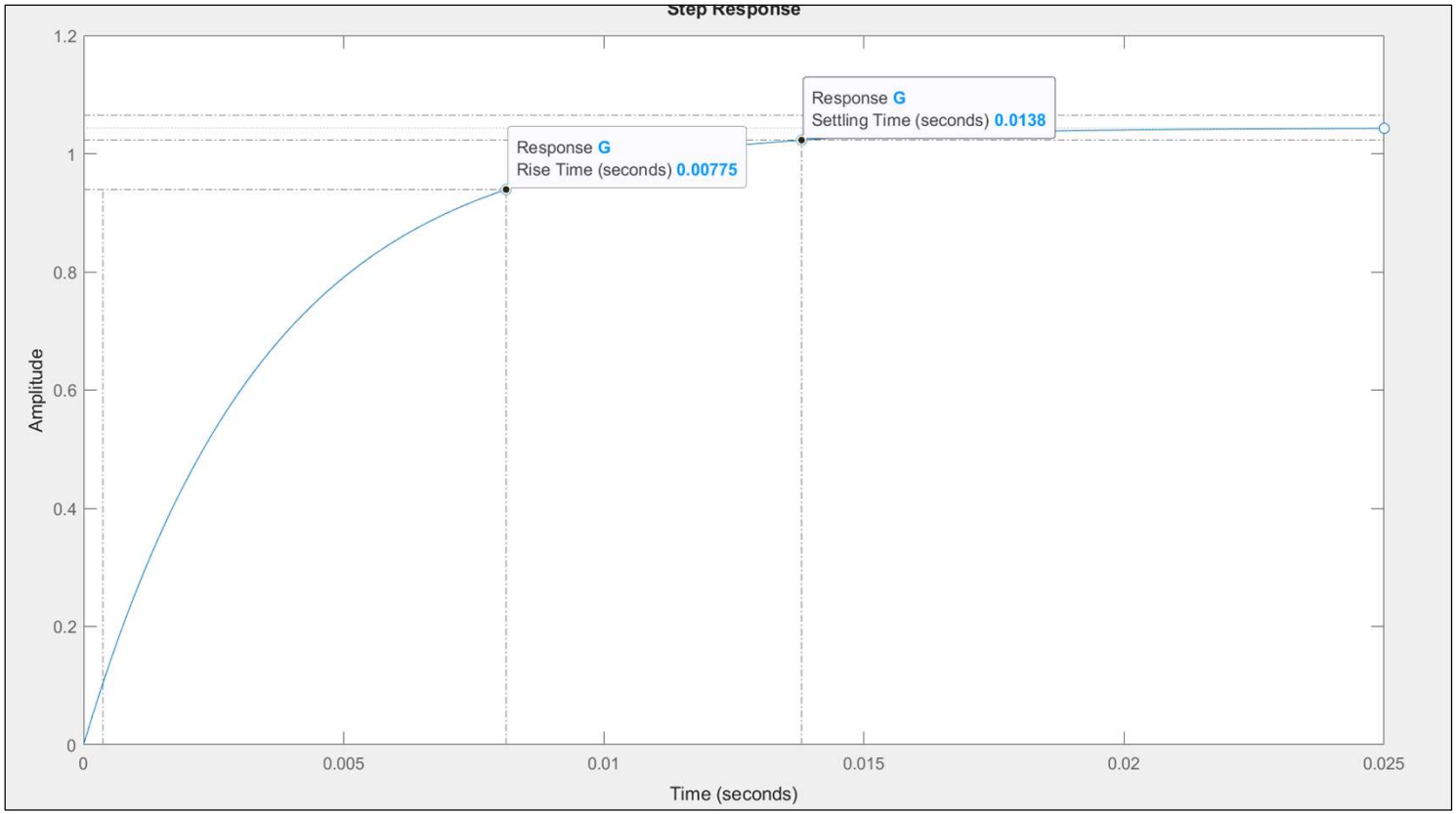
$$Pressure = \frac{Force}{Area}$$

Plant Transfer Function (P(s)): The system is assumed to be the second-order plant derived from the linearization:

$$P(s) = \frac{Y(s)}{U(s)} = \frac{1.076 \times 10^{11}}{s^2 + 3.637 \times 10^8 s + 1.031 \times 10^{11}}$$

$$\omega_n = \sqrt{1.031 \times 10^{11}} \approx \mathbf{3.211 \times 10^5} \text{ rad/s}$$

$$\zeta = \frac{3.637 \times 10^8}{2\omega_n} \approx \frac{3.637 \times 10^8}{2(3.211 \times 10^5)} \approx \mathbf{566.6}$$

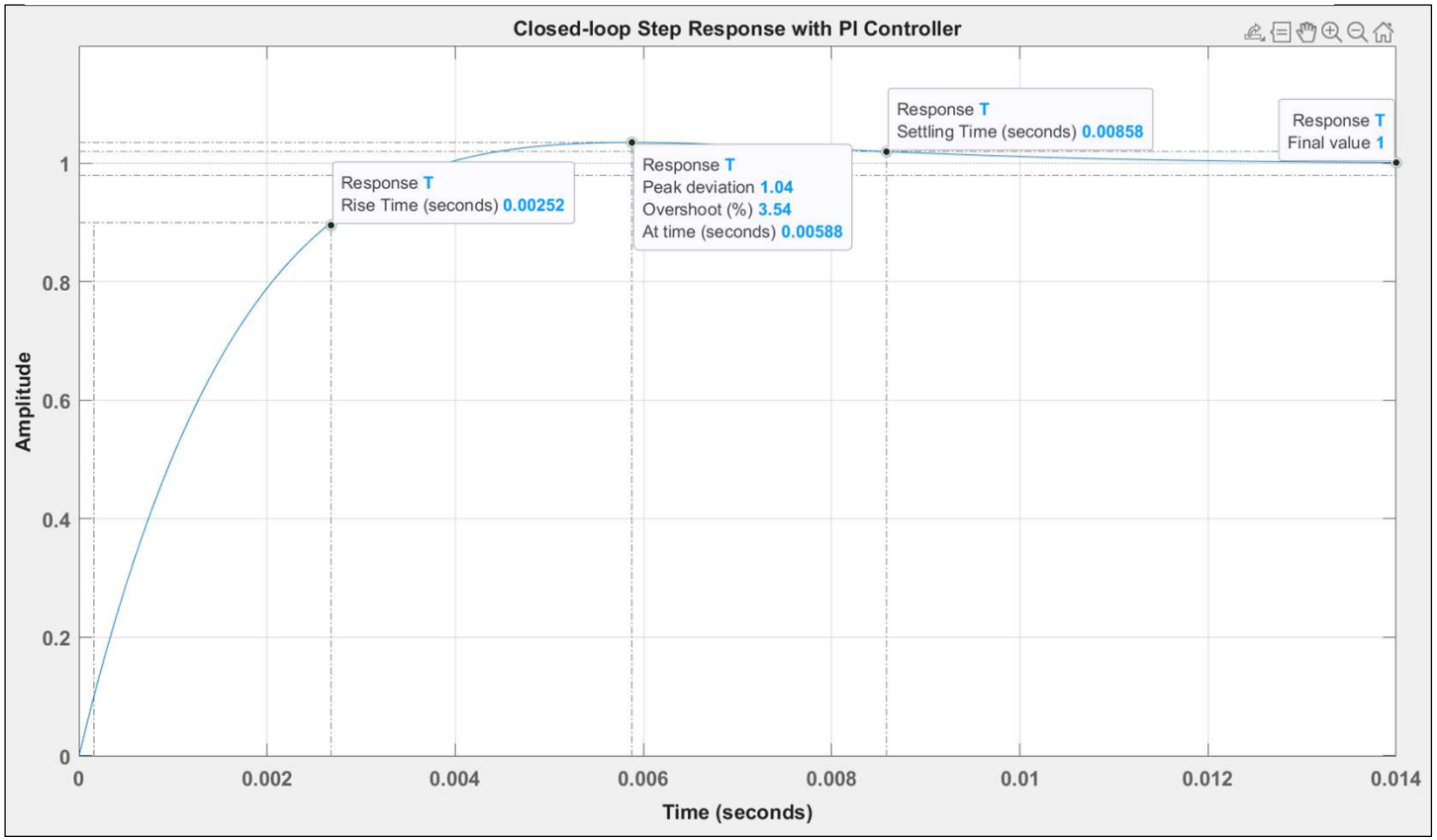


$$K = \frac{1.076 \times 10^{11}}{1.031 \times 10^{11}} \approx \mathbf{1.044}$$

Closed Loop response of the plant

The closed-loop transfer function is:

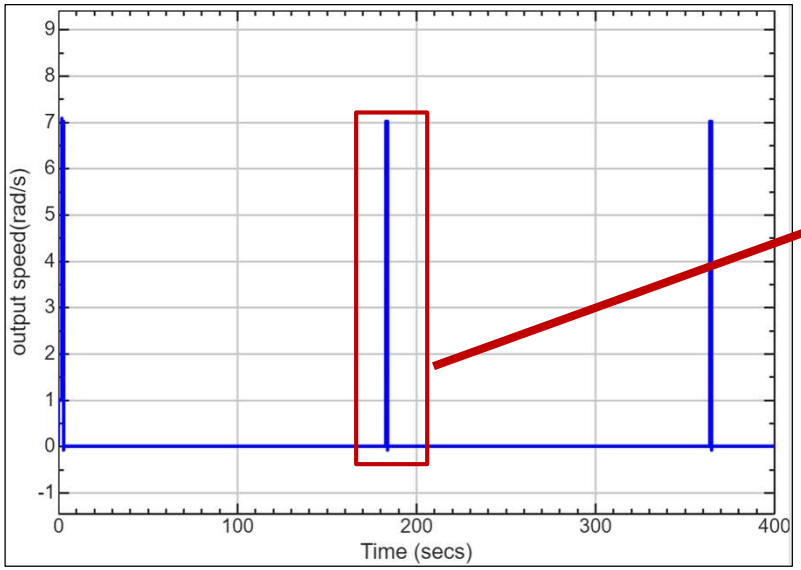
$$G_{CL}(s) = \frac{N_{OL}(s)}{D_{CL}(s)} = \frac{(2.39 \times 10^{11})s + (9.90 \times 10^{13})}{s^3 + (3.64 \times 10^{8})s^2 + (3.42 \times 10^{11})s + 9.90 \times 10^{13}}$$

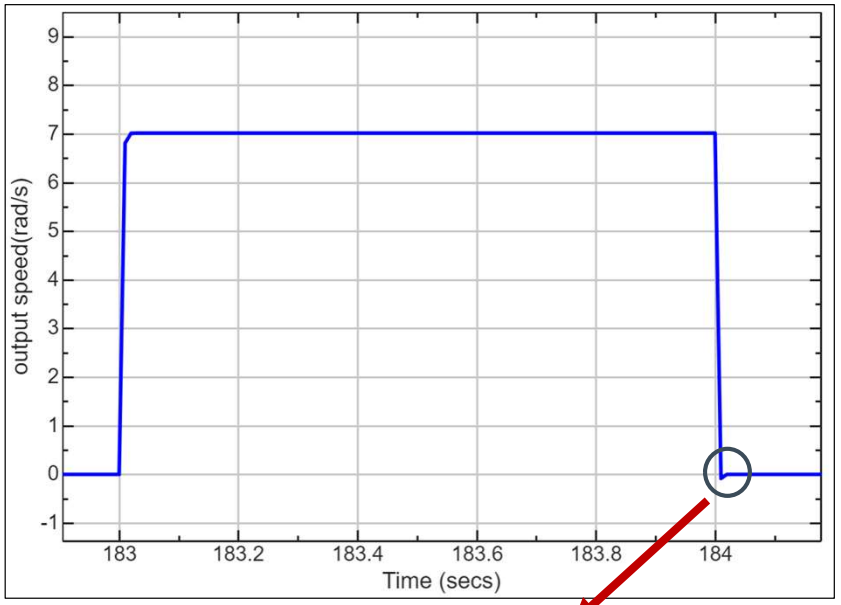


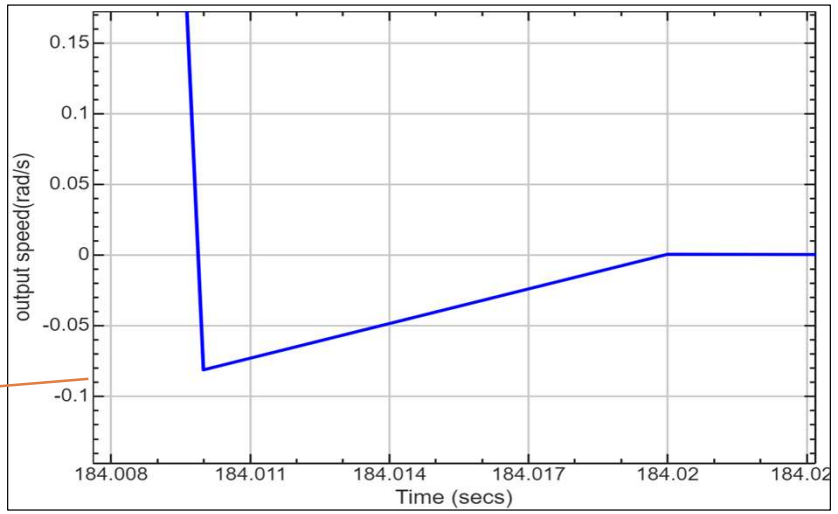
The optimized gain parameters are:
P =2.22, I =919.6213

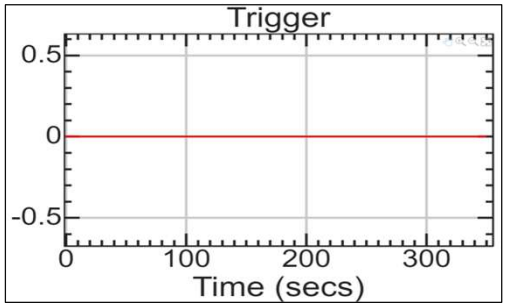
Simulation Results:

Motor speed(rad/s) Vs Time(s): Basal







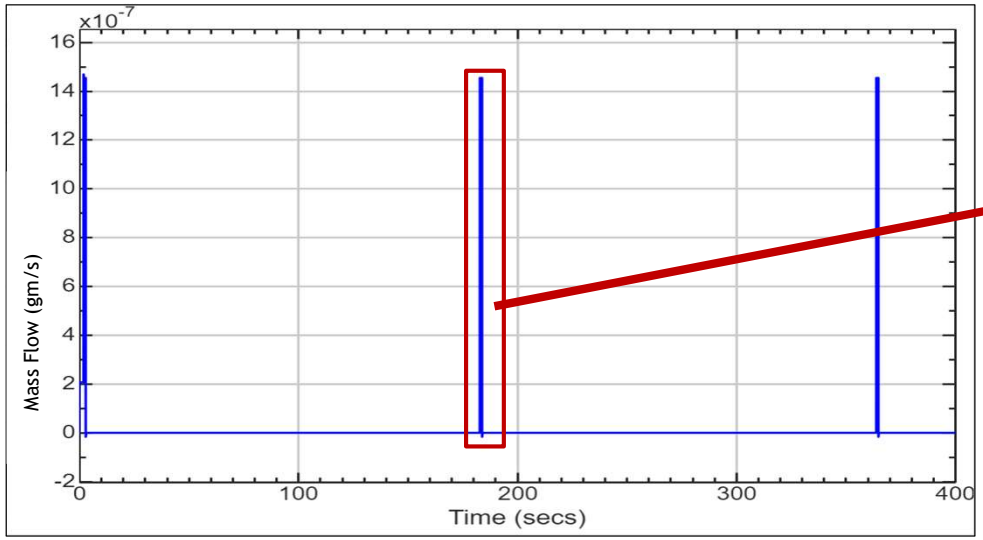


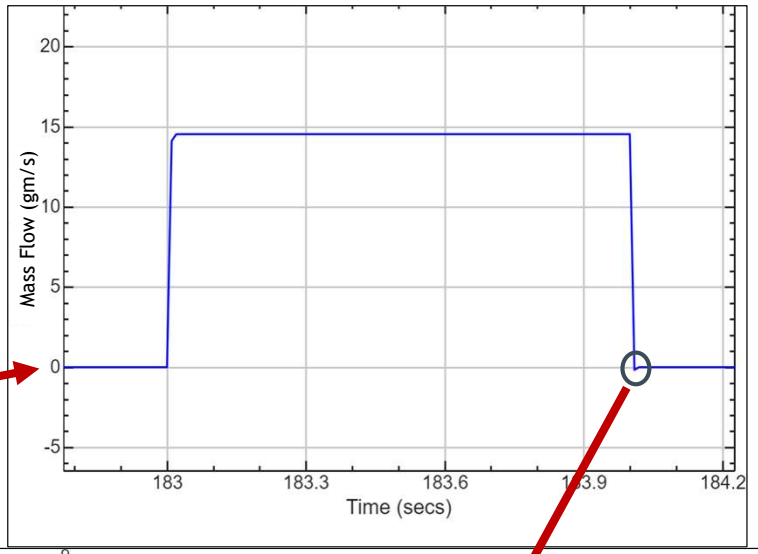
- 0.09

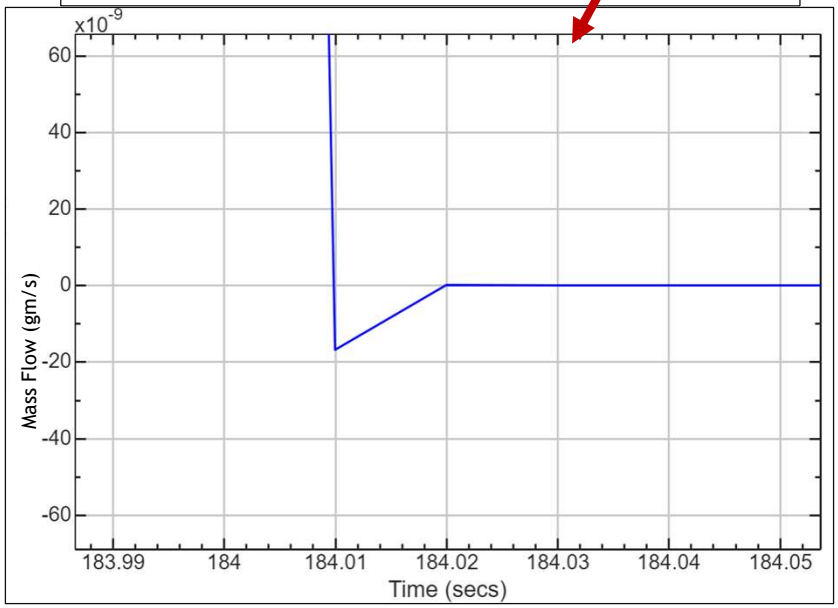
Undershoot

Simulation Results:

Mass flow Vs Time(s): Basal







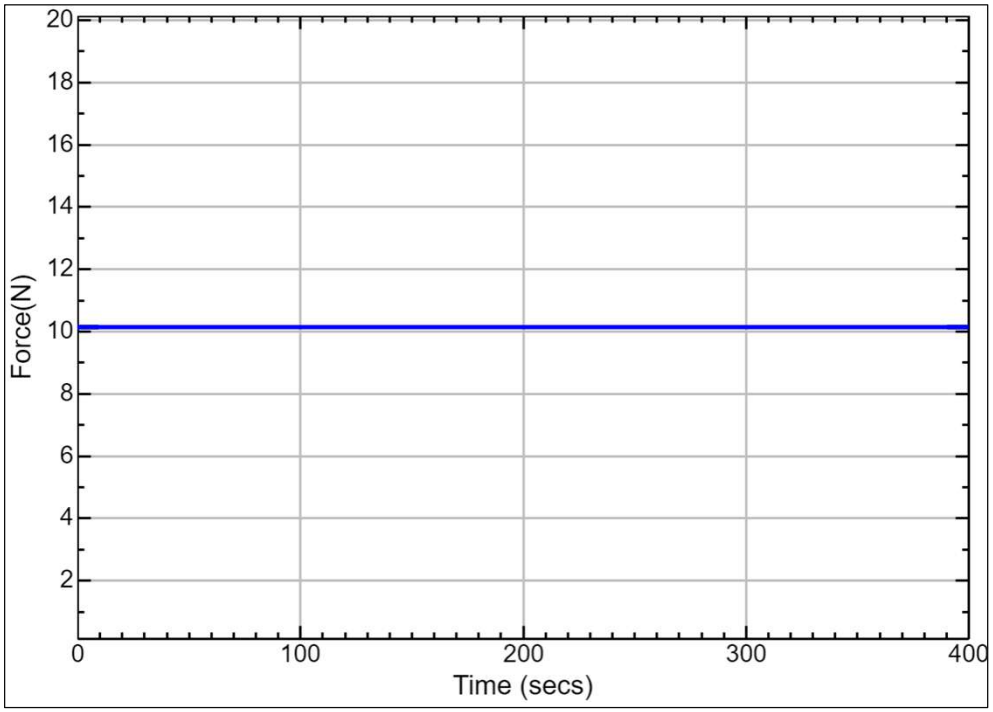
24 Hour avg. = 19.94IU

To be Delivered = 20IU

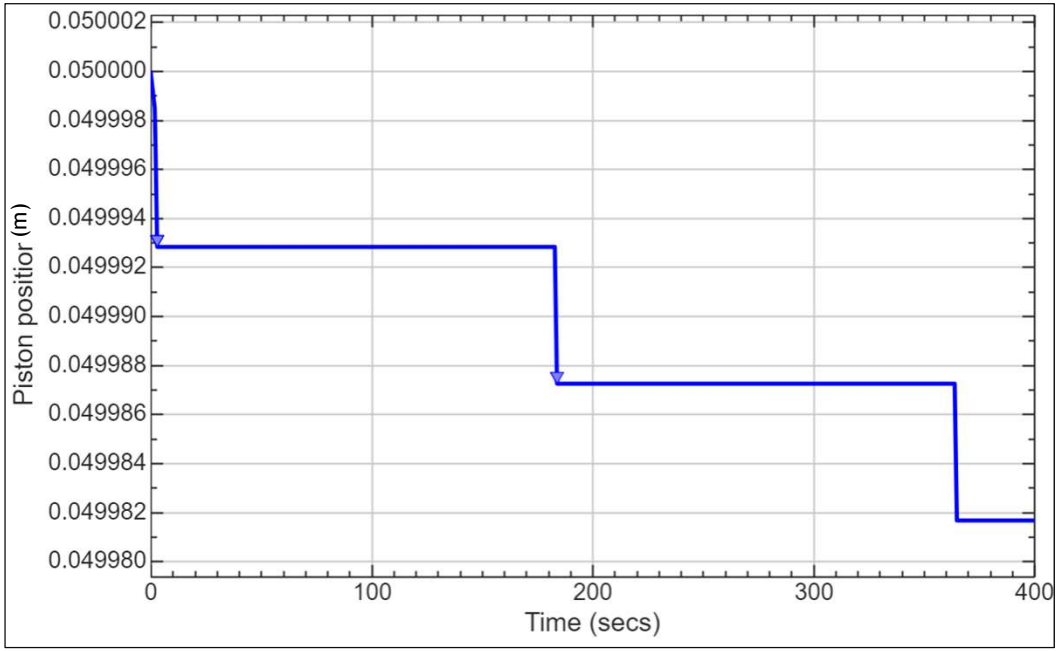
99.7% Accuracy

Simulation Results:

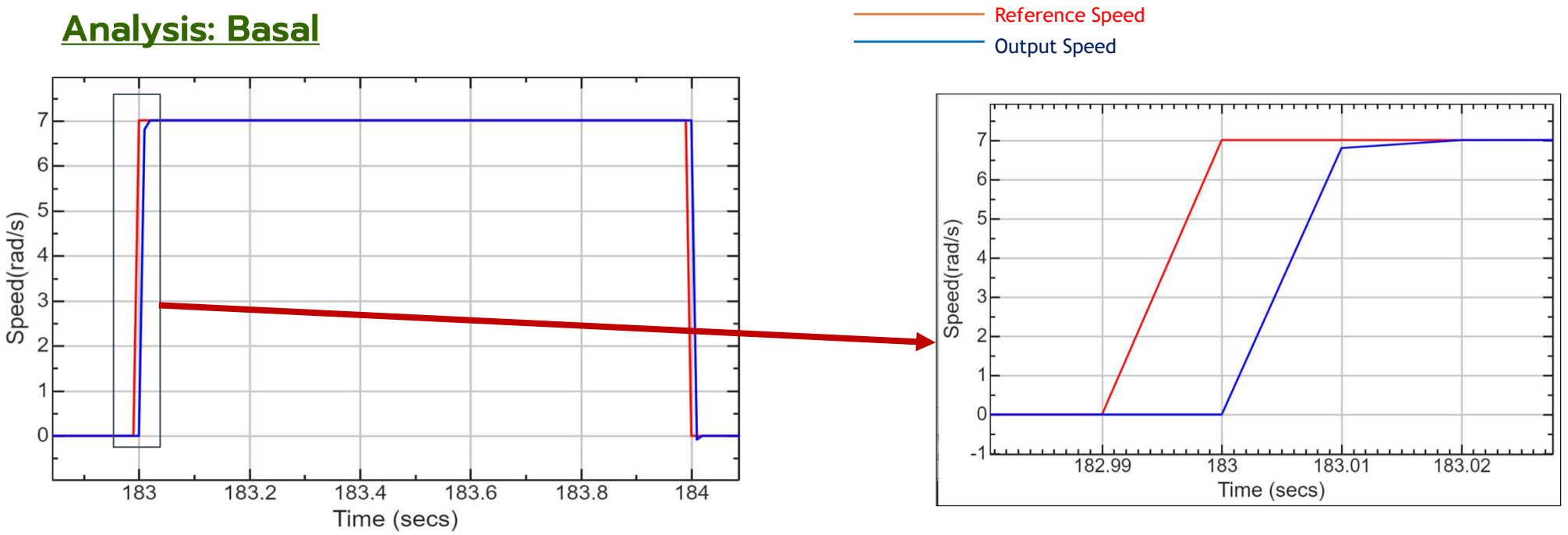
Force(N) Vs Time(s): Basal



Piston Position(m) Vs Time(s): Basal



Analysis: Basal

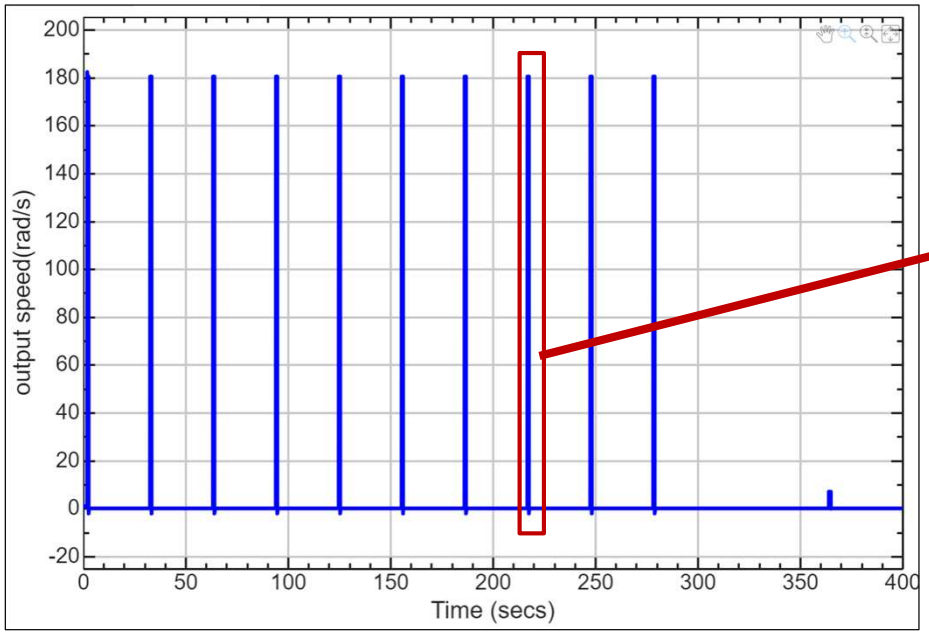


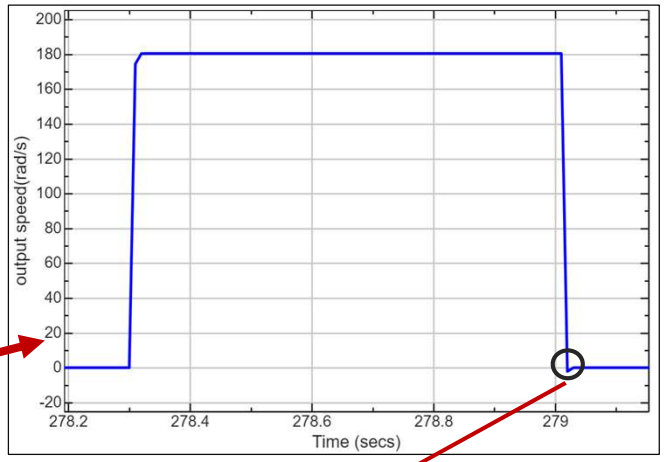
- **Reference Speed:** 7 rad/s
- **Peak Actual Speed:** 7 rad/s
- **Percent Overshoot: ≈ 0 %**
- **Steady-State Value:** 7 rad/s
- **Steady-State Error: ≈ 0 (negligible)**

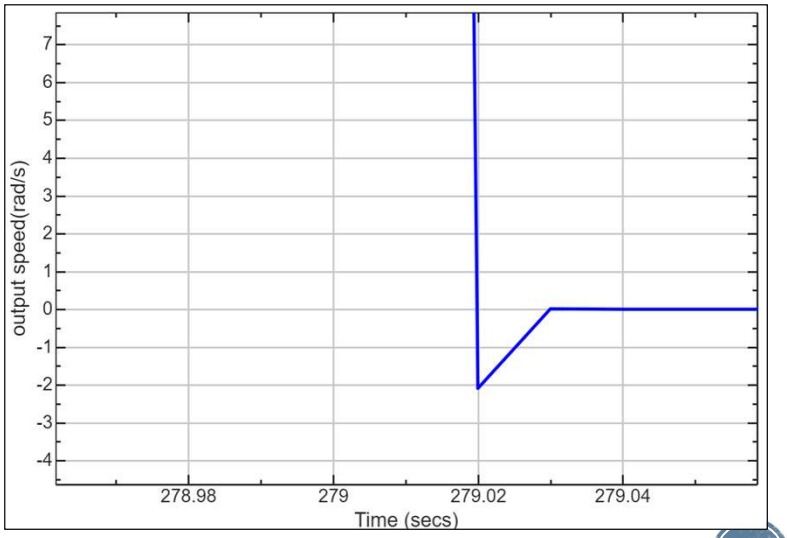
- **Delay Time (Td): ≈ 0.01 s**
- **Rise Time (Tr): ≈ 0.015–0.016 s**
- **Settling Time (Ts): ≈ 0.03–0.04 s**
- **Response Characteristics:**
  Very fast rise, Minimal overshoot, Smooth settling and good tracking accuracy

Simulation Results:

Motor speed(rad/s) Vs Time(s): Bolus

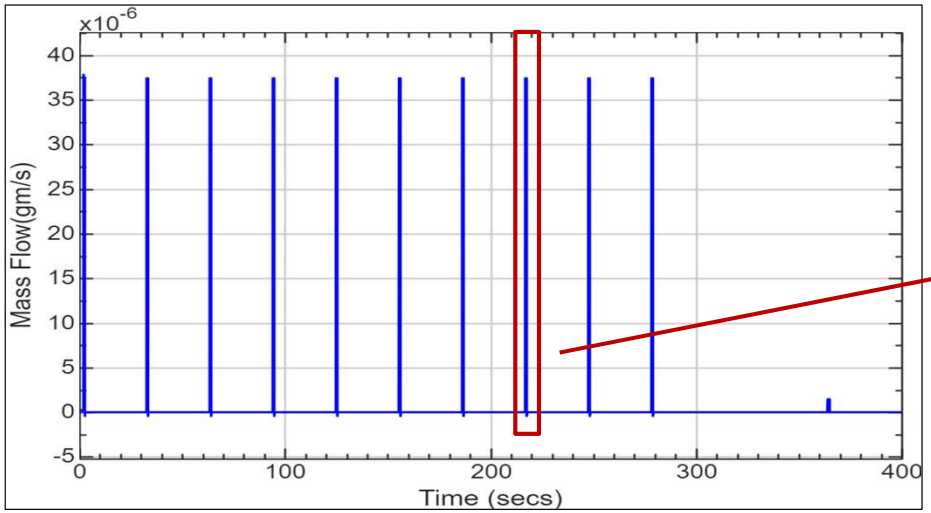


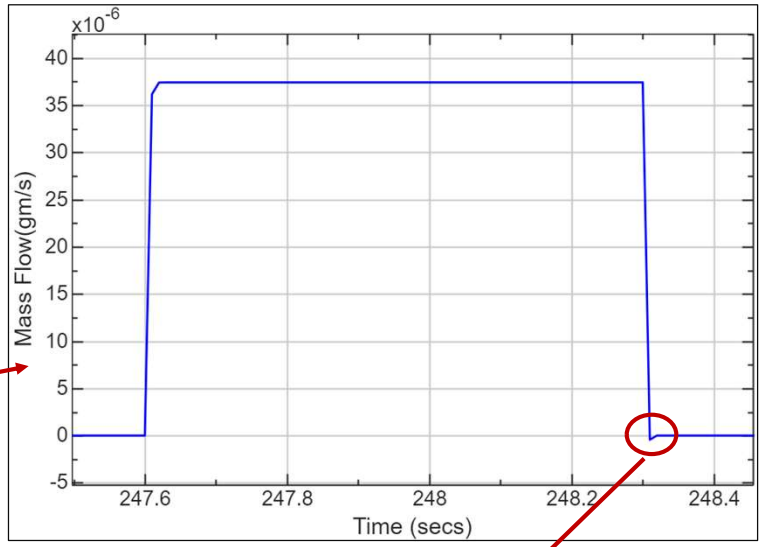


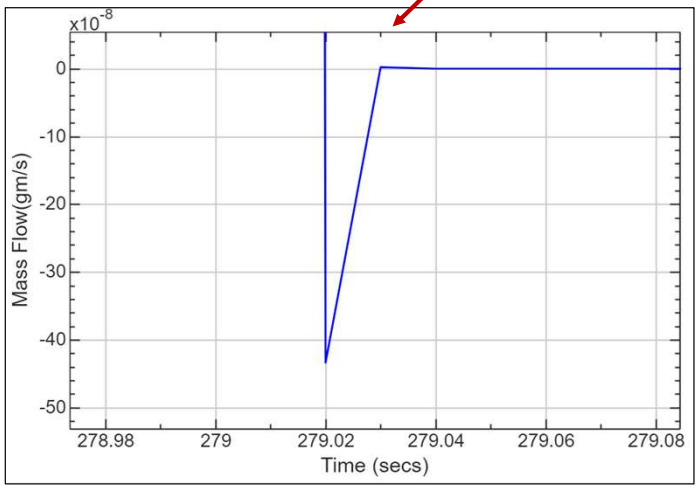




Mass Flow(gm/s) Vs Time(s): Bolus





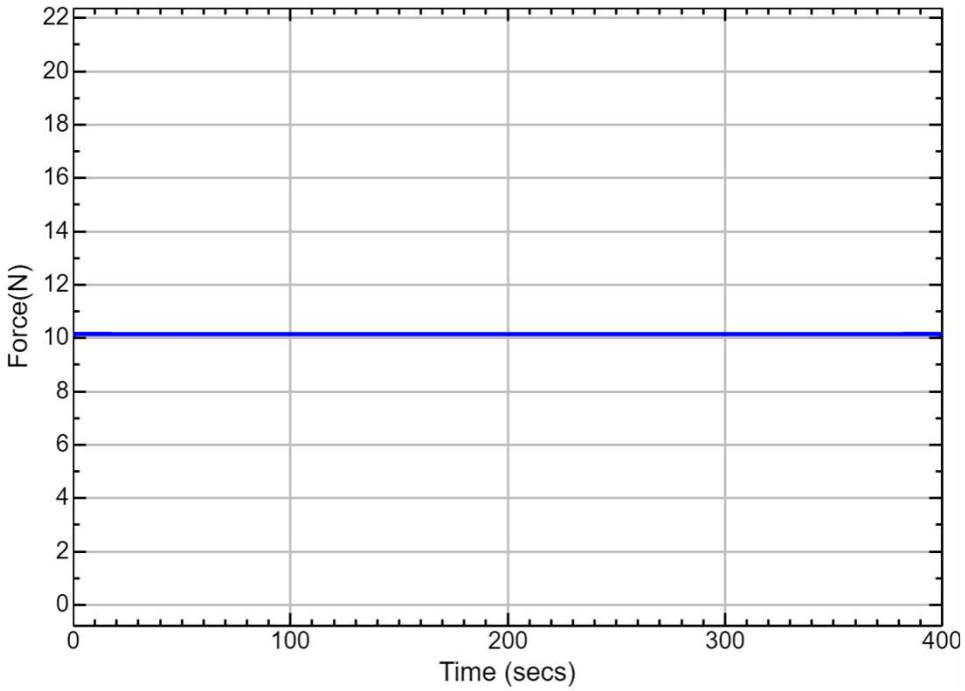


5 Min avg. = 7.58IU

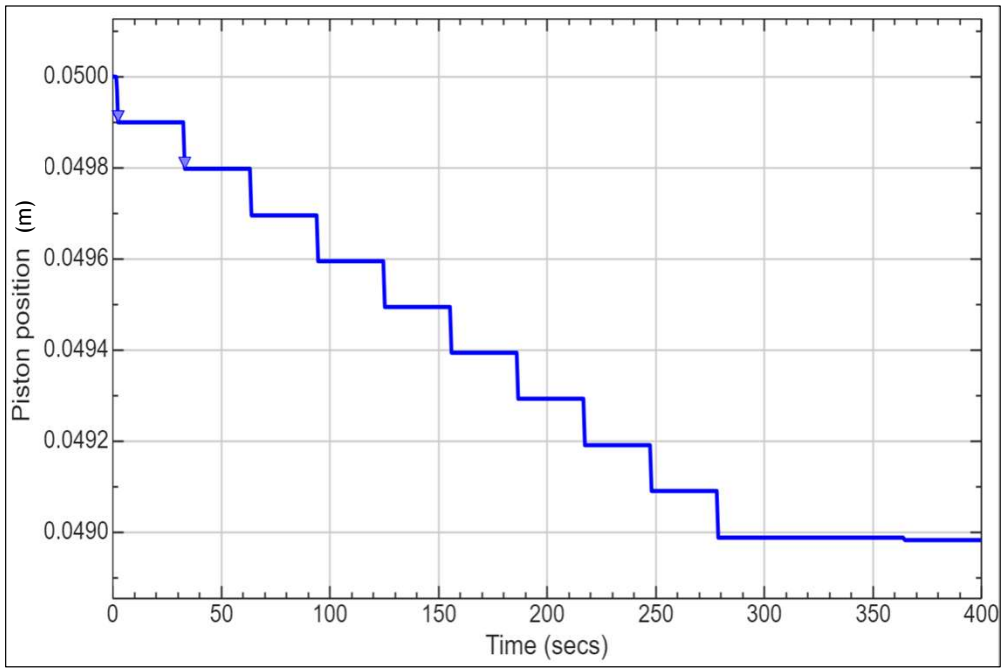
To be Delivered = 7.378IU

99.97% Accuracy

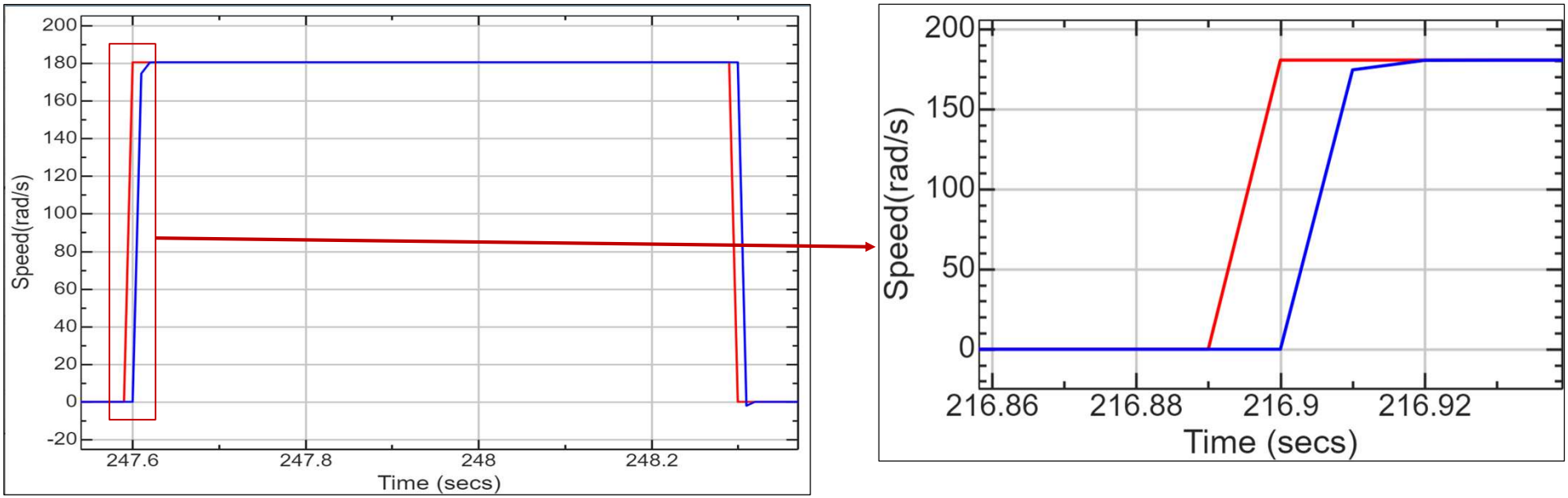
Force(N) Vs Time(s): Bolus



Piston Position(m) Vs Time(s): Bolus



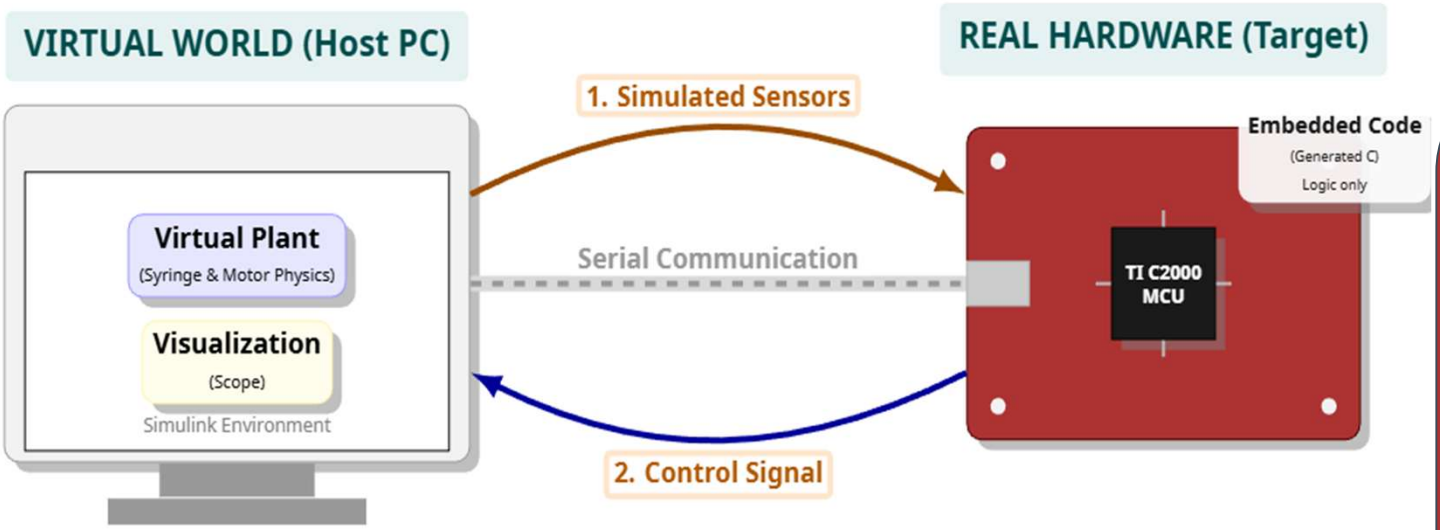
Analysis: Bolus



- **Reference Speed:** 180 rad/s
- **Peak Actual Speed:** ~180 rad/s
- **Percent Overshoot: ≈ 0 %**
- **Steady-State Speed:** 180 rad/s
- **Steady-State Error: ≈ 0 rad/s (negligible)**
- **Rise Time (Tr): ≈ 0.020–0.022s**
- **Delay Time (Td): ≈ 0.012–0.015 s**
- **Settling Time (Ts): ≈ 0.04–0.05 s**
- **Response Type: Fast, stable, minimal overshoot, good tracking accuracy**

Hardware In Loop

Real-Time Control Verification using TI C2000 & Simulink



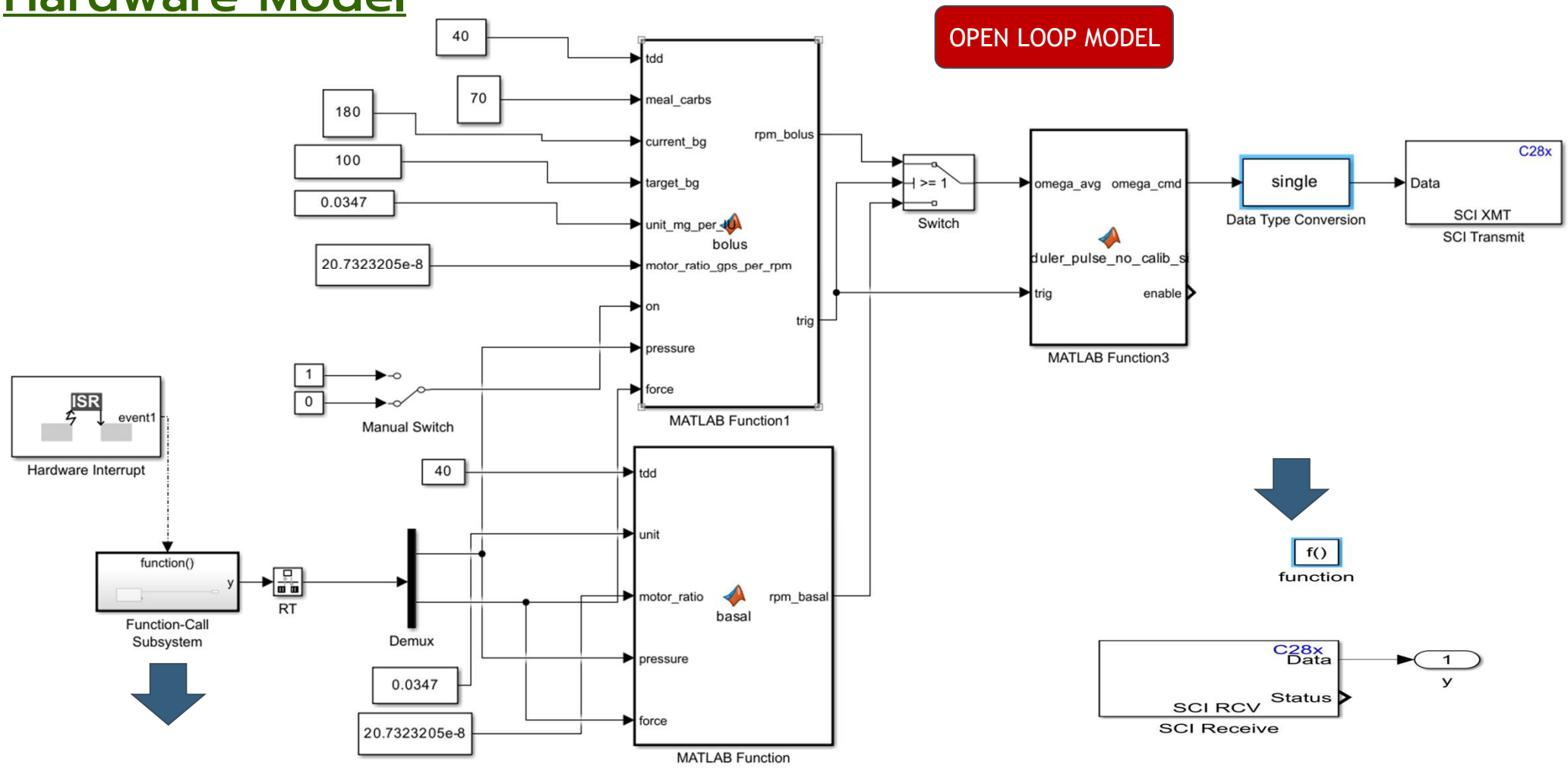
How it works:

- **Plant Simulation:** The physics is calculated mathematically in Simulink (Virtual).
- **Real-Time Control:** The TI C2000 executes the actual algorithms.
- **The Loop:** Physical signals are exchanged in real-time. If the code fails, the virtual model breaks, not the real one.

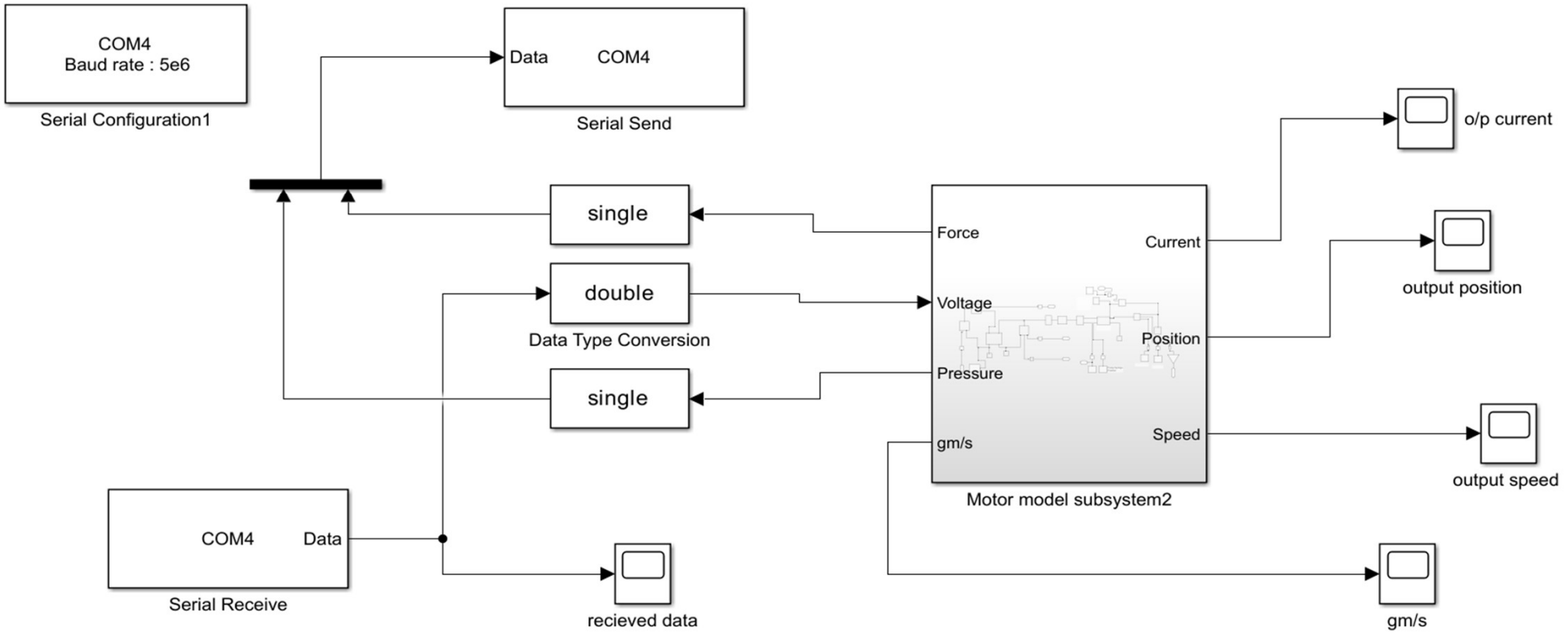
TI C2000 F2837x MCU

- ❑ **176-pin** high-performance real-time control MCU (F28379D variant).
- ❑ Dual **C28x CPUs** + **CLA** accelerator for parallel control tasks.
- ❑ High-resolution **PWM**, fast **12-bit/16-bit ADCs**, **DAC**, **eCAP**, **eQEP**.
- ❑ Multiple comms: **SPI, I2C, UART, CAN, USB**

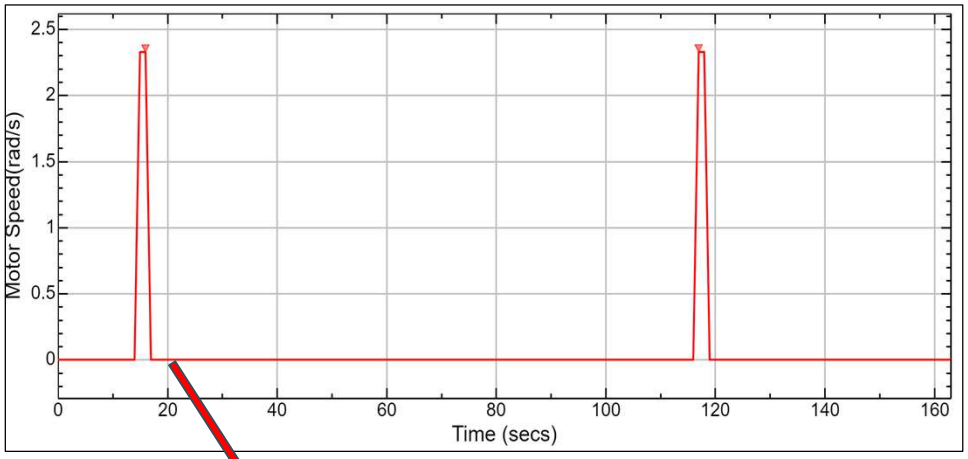
Hardware Model



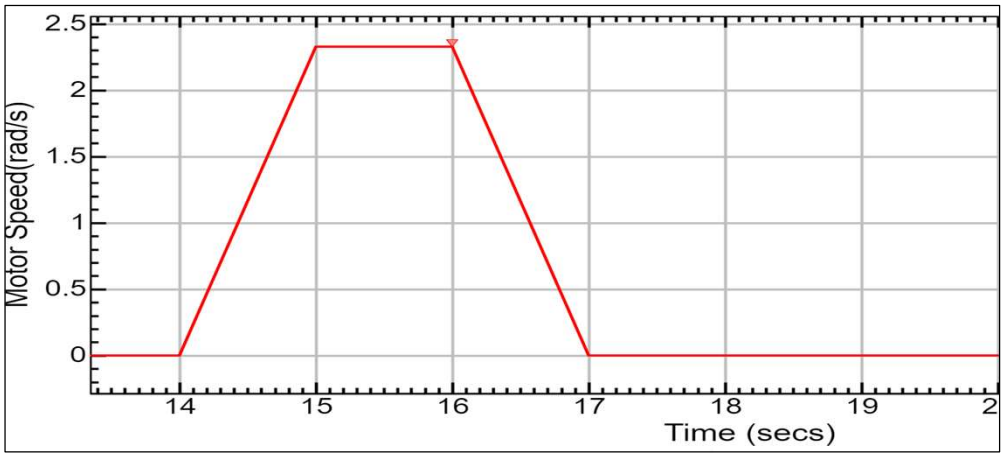
Host Simulink Model



Simulation Results

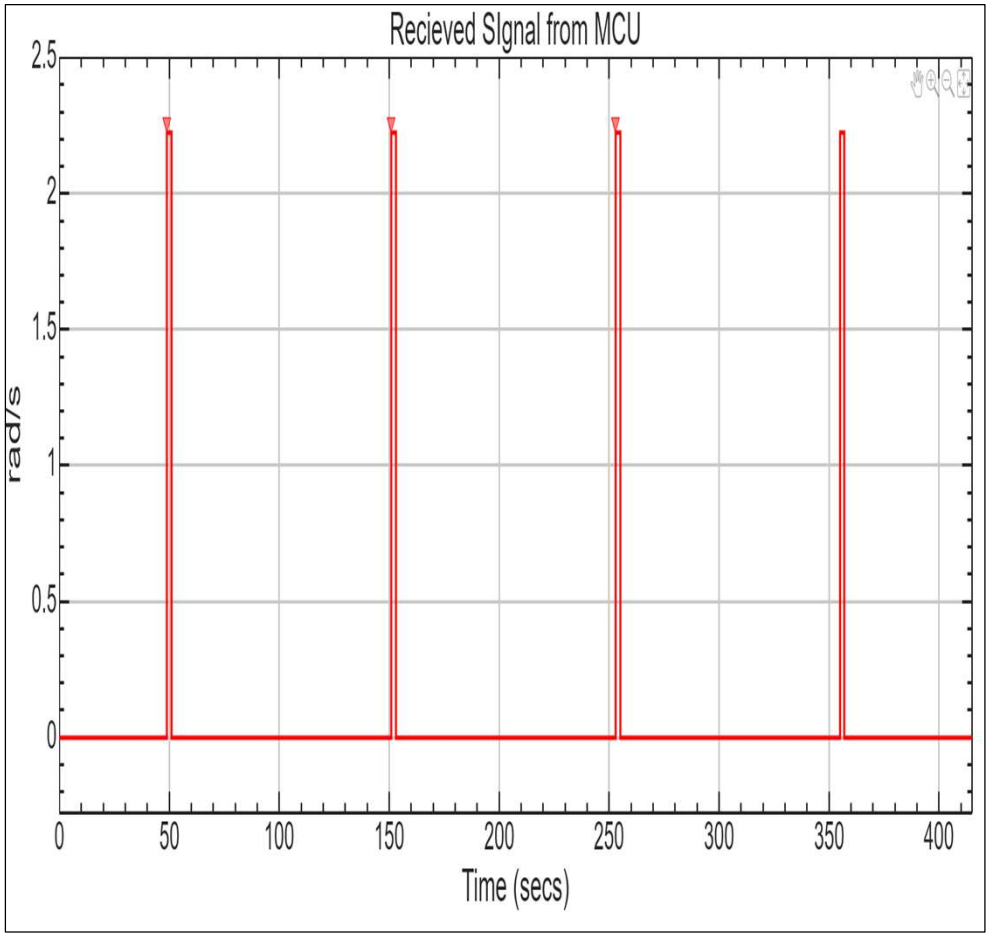


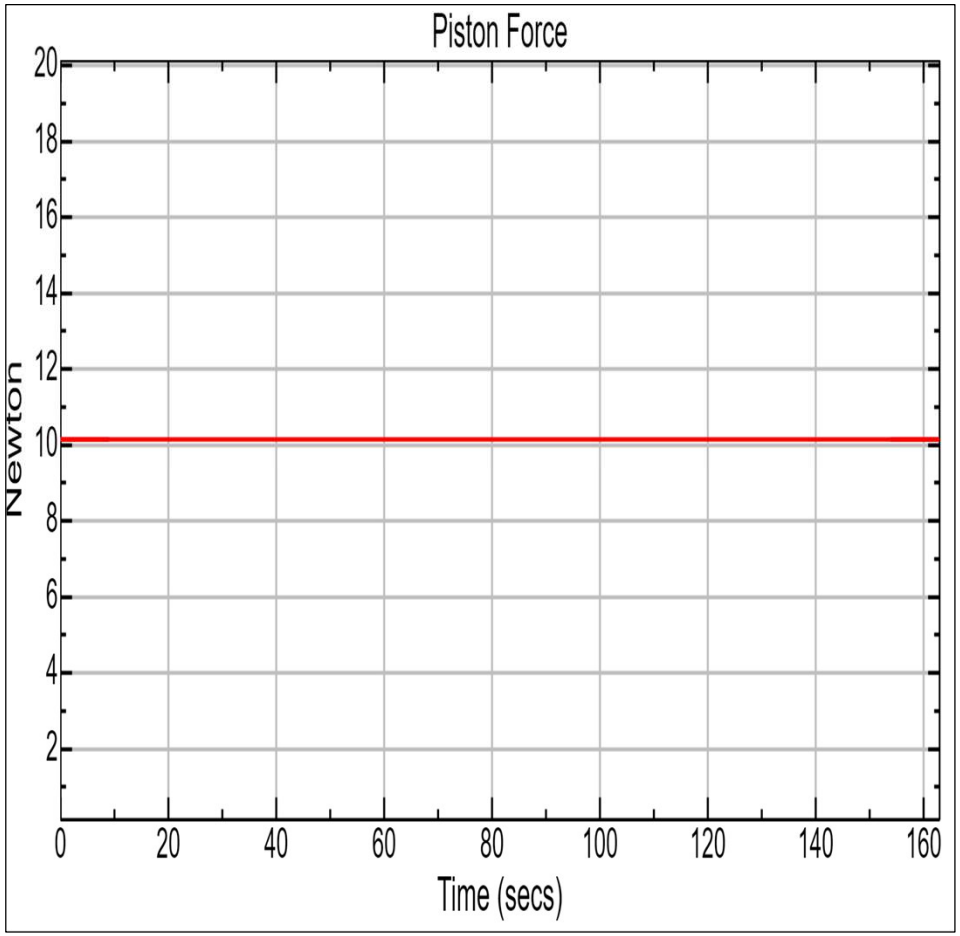






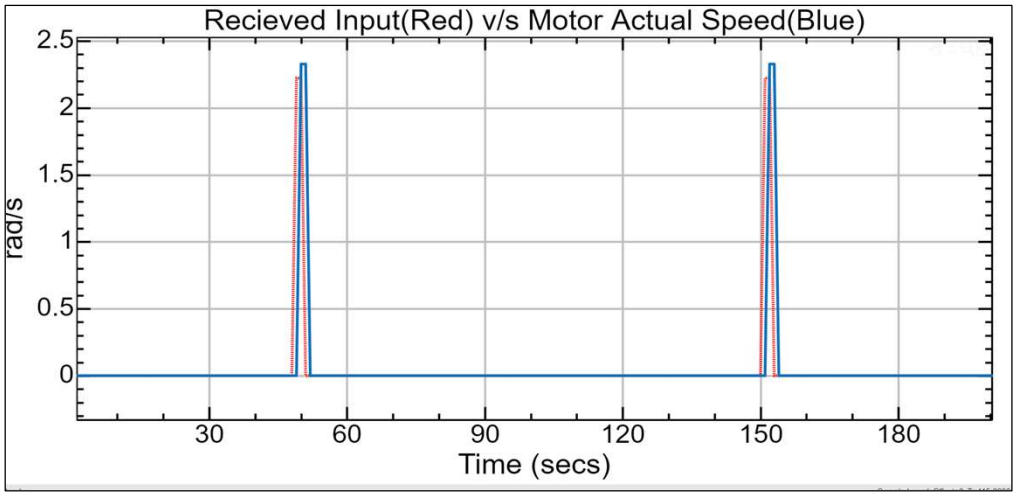
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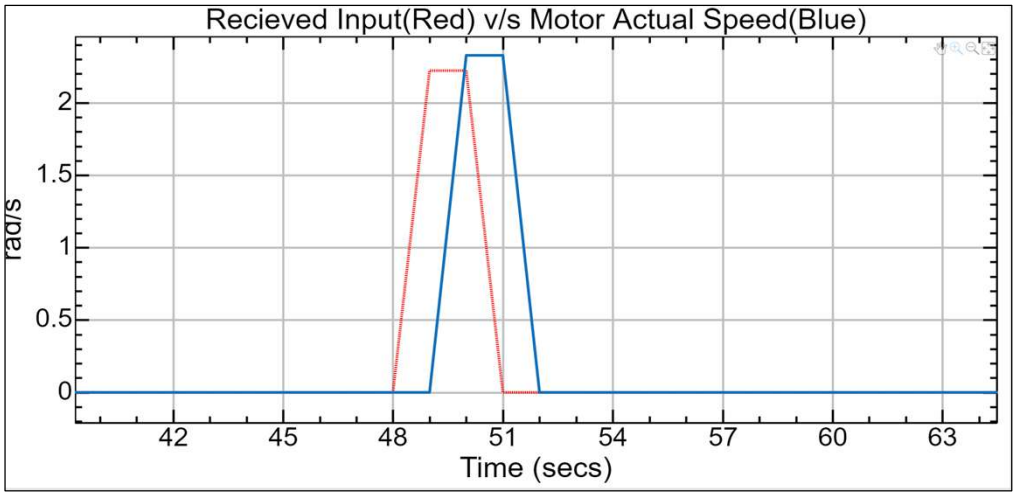




Analysis

Reference Speed
Output Speed





- **Reference Speed:** 2.20 rad/s
- **Peak Speed:** 2.32 rad/s
- **Percent Overshoot: 5.45%**
- **Steady-State Speed:** 2.32 rad/s
- **Steady-State Error:** 0.08 rad/s (**5.45%**)
- **Rise Time (Tr):** 1.2 s
- **Delay Time (Td):** 1 s
- **Settling Time (Ts):** 1.4 s

Conclusion: Democratizing Insulin Pump Therapy

Why This Design Matters

This project demonstrates that clinical-grade insulin pump therapy need not be financially prohibitive. Our target is to achieve cost reduction without compromising safety or efficacy.

Scalability vision: Modular design enables regional manufacturing with locally sourced components, reducing supply chain dependencies and enabling sub.

Future Development:

Closed-Loop Improvements

- Implement advanced control methods for better robustness.
- Develop nonlinear system modeling for improved real-world accuracy.
- Add adaptive gain scheduling to optimize controller performance.

Software & Embedded Enhancements

- Further optimize Simulink-generated code for TI C2000 efficiency.
- Integrate advanced safety logic (occlusion detection, predictive monitoring).
- Implement automatic sensor calibration to reduce long-term drift.

System Expansion & Connectivity

- Add Bluetooth integration with smartphone or smartwatch applications.
- Develop custom PCB and final hardware casing with charging support.
- Provide user interface for manual override, monitoring, and control.

Thank You

